Jiya Kaul

Assessment: Technical Product Manager | FactWise

Submission File: Jiya Kaul TPM.pdf
Facility Scale: 74 Beds (Full Occupancy)
Working Prototype & Deployed Link Included

Problem Understanding: A virus quarantine facility (74 beds, 15% expected mortality benchmark) is experiencing severe staff burnout. Patients are considered cured after 3 full days without a fever. Currently, lack of workflow coordination means doctors visit before nurses log vitals, recovered patients stay an extra 1–2 days waiting for manual discharge handoffs, and leadership has no way to track whether the center is meeting the 85% success benchmark.

Q1) In your app's first version, what are the different kinds of users, and which features and actions are relevant for which user types?

To solve the head doctor's bottlenecks immediately, I defined 4 user types in Version 1. Each user gets a dedicated workspace so responsibilities and clinical sequences stay clear:

User Type	Operational Role & Bottleneck Solved	Features & Permitted Actions
1. Nurse	Owns the daily morning vitals check. Stops paper journal logging, prevents duplicate measurements, and ensures no bed is missed.	• **Shift Queue Tabs:** Toggle between *Needs Temp Check Today* and *Completed Today*. • **Log Temperature Modal:** Fast input in °F with instant color feedback (Green: <99.5°, Orange: 99.5–100.3°, Red: ≥100.4°F). • **Fever Streak Tracker:** Real-time badge tracking consecutive fever-free days (e.g. "2 / 3 Days"). • *Actions:* Record daily vitals and clinical notes. Cannot approve discharges or alter bed allocations.
2. Attending Doctor	Examines patients after vitals are ready, reviews trendlines, prescribes care, and certifies patient cure without manual admin lag.	• **Prioritized Rounding Queue:** Clearly splits patients into *Ready for Round* (vitals logged) vs *Pending Vitals* (amber warning indicator). • **Vitals Trendline Sparkline:** 7-day temperature curve with a 100.4°F fever reference line. • **Doctor Notes Log:** Document clinical evaluations and treatment changes. • *Actions:* 1-click discharge sign-off (gated until 3 fever-free days are met). Fallback: record vitals if urgent.
3. Facility Admin	Manages physical bed turnover across all 74 beds, processes new admissions, and vacates beds immediately upon patient departure.	• **74-Bed Interactive Matrix:** Visual tiles for Beds 1–74 color-coded by real-time status (Red: Occupied, Amber: Discharge Pending, Green: Available). • **Discharge Desk:** Instant queue of doctor-approved discharges; 1-click "Finalize Discharge & Release Bed". • *Actions:* Admit patients into verified green beds; finalize checkout; log patient mortality. Cannot edit medical vitals.
4. Head Doctor / Medical Director	Monitors clinical quality benchmarks, mortality rates, and daily staff operational compliance across the entire facility.	• **Quality Scorecards:** Live Mortality Rate vs the 15% benchmark and Cure Rate vs the 85% target. • **Critical Alert Banner:** Flashing banner if mortality exceeds 15.0%. • **Shift Progress Telemetry:** Progress bars tracking Nurse Checks % and Doctor Rounds % completed today.

User Type	Operational Role & Bottleneck Solved	Features & Permitted Actions
		• *Actions:* High-level oversight, shift audit, and Average Length of Stay (ALOS) tracking.

Q2) Functional Prototype (Built & Deployed)

I built and deployed a production web application so the complete clinical workflow can be tested live:

- **Live Deployed Working Link:** https://fact-wise-assessment-two.vercel.app
- **GitHub Repository:** https://github.com/Jiyakaul/FactWise-assessment
- **Tech Stack:** Next.js 14 App Router, React 18, TypeScript, Tailwind CSS, Recharts, Lucide Icons.

Demonstrable Screens and Workflows in the Prototype:

1. **Screen 1: Nurse Station (Temperature Tracking)**
   Shows active patients sorted by bed number. Includes search and tab filtering. *Try Workflow:* Find **Bed 5, 12, or 28** (pre-loaded with exactly 2 fever-free days). Click "Record Temp", enter a normal reading (e.g. 98.6°F), and submit. The streak advances to 3/3 and status updates to "Eligible for Discharge".
2. **Screen 2: Doctor Rounds & Treatment**
   Separates patients into *Ready for Round* and *Pending Nurse Vitals* (with ⚠️ warnings). *Try Workflow:* Select **Bed 40** (pre-loaded with 3 fever-free days). Opening the modal displays the 7-day temperature trendline and activates the glowing green "Sign Discharge Order" button.
3. **Screen 3: Admin & Bed Management (74-Bed Board)**
   Displays all 74 beds as color-coded visual tiles. *Try Workflow:* Go to the "Discharge Desk" section where **Bed 55** is pre-approved by the doctor. Click "Finalize Discharge & Release Bed"—Bed 55 immediately turns green (Available) in the 74-bed matrix.
4. **Screen 4: Head Doctor Analytics Dashboard**
   Computes live mortality against the 15% benchmark, cure rate against 85%, and shows shift compliance gauges.

Q3) Additional Logical Constraints for Developers

To prevent real-world medical errors in an understaffed ward, I documented these non-obvious logic rules for the engineering team:

1. Sequential Dependency (Nurse-First Rule)

- Daily vitals flags (`todayTempRecorded`, `todayDoctorVisited`) reset automatically at `00:00:00`.
- If an Attending Doctor opens a patient whose nurse check is still pending, display an amber warning: "*Warning: Temperature has not been recorded today by a nurse.*"
- If the doctor takes the vitals themselves, the intake logs it with `recordedBy: "doctor"` and clears the nurse queue item so duplicate readings are avoided.

2. The Strict 3-Day Fever-Free Cure Algorithm

- **Fever Line:** Any reading $T \ge 100.4^\circ\text{F}$ (38.0°C) is flagged as fever.
- **Strict Reset Rule:** If a patient has 2 fever-free days and registers 100.4°F on Day 3, the counter **strictly resets to Day 0**. No grandfathering.
- **Discharge Gate:** The "Sign Discharge Order" button is hard-disabled until `consecutiveFeverFreeDays ≥ 3`.
- **Two-Stage Handoff:** Doctor approval flags status as `doctor_approved_discharge` (alerting Admin desk). The bed remains occupied until Admin completes the checkout, ensuring front desk is never out of sync.

3. Capacity, Input Validation & Metric Integrity

- **74-Bed Ceiling:** Hard stop at 74 active patients. Intake form is disabled when active patients $= 74$.
- **Physiological Bounds:** Temperature inputs must be validated within $90.0^\circ\text{F} \le T \le 110.0^\circ\text{F}$.
- **Closed Cases Formula:** To avoid distorting mortality with patients still under treatment, calculate rates on closed outcomes: $$\text{Mortality Rate} = \frac{\text{Total Deceased}}{\text{Discharged Cured} + \text{Deceased}} \times 100, \quad \text{Cure Rate} = \frac{\text{Discharged Cured}}{\text{Discharged Cured} + \text{Deceased}} \times 100$$
- **Alert Threshold:** If Mortality Rate > 15.0%, trigger an emergency banner across leadership screens.

Q4) Sprint Planning Mathematical Optimization (Knapsack Solution)

Sprint Parameters: Velocity = 25 Story Points (SP). Already committed: 3 stories × 5 SP = 15 SP.
Remaining Available Capacity = 10 SP.
Value Rules: Low Priority = 1 unit ($V_L = 1$), Medium = 3 units ($V_M = 3$), High = 6 units ($V_H = 6$).

Feature	Priority	Value (V)	Effort (W)	Value Density (V/W)	Sprint Decision
J	Low	1	0.1	**10.00**	**INCLUDE (Highest density)**
A	High	6	1.0	**6.00**	**INCLUDE (High value, low effort)**
I	Low	1	0.3	**3.33**	**INCLUDE**
F	Medium	3	1.0	**3.00**	**INCLUDE**
G	Low	1	0.4	**2.50**	**INCLUDE**

Feature	Priority	Value (V)	Effort (W)	Value Density (V/W)	Sprint Decision
B	High	6	3.0	**2.00**	**INCLUDE**
H	Low	1	0.5	**2.00**	**INCLUDE**
D	Medium	3	2.0	**1.50**	**INCLUDE**
C	High	6	5.0	1.20	EXCLUDE (Consumes 50% of budget for 6 val)
E	Medium	3	3.0	1.00	EXCLUDE (Lowest density)

Comparing the Two Competing Options:

1. **Option 1 (Intuitive "High Priority First" Approach):** Select High features {A, B, C} (9 SP, 18 Value) + F (1 SP, 3 Value).
   • **Total Effort:** $1.0 + 3.0 + 5.0 + 1.0 = \mathbf{10.0\text{ SP}}$ (Zero buffer)
   • **Total Value:** $6 + 6 + 6 + 3 = \mathbf{21\text{ Value Units}}$
2. **Option 2 (Knapsack Optimization - Maximum Value & Minimum Effort):** Notice Feature C consumes 5 SP for only 6 value (density 1.2). If we omit C and E, we can include High features {A, B}, Medium features {D, F}, and all Low features {G, H, I, J}:
   • **Total Effort:** $1.0 + 3.0 + 2.0 + 1.0 + 0.4 + 0.5 + 0.3 + 0.1 = \mathbf{8.3\text{ Story Points}}$
   • **Total Value:** $6 + 6 + 3 + 3 + 1 + 1 + 1 + 1 = \mathbf{22\text{ Value Units}}$
   • **Safety Buffer Remaining:** $10.0 - 8.3 = \mathbf{1.7\text{ Story Points}}$

> **My Recommendation as TPM:** Add **Features A, B, D, F, G, H, I, and J**.
> This mathematically maximizes delivered value at **22 points** (beating the intuitive 21 points) while consuming only **8.3 story points**. This leaves a **1.7-point safety buffer (17% contingency)** to protect sprint commitments against unexpected hospital emergencies.

Extra: Out-of-the-Box Product Ideas & Security Roadmap (v2 Scope)

To build beyond the basic MVP and make the facility truly robust, here are 4 high-impact product ideas I would prioritize:

1. Bedside QR / Barcode Wristband Scanning (Patient Safety)

In an understaffed ward, nurses can easily confuse patients in neighboring beds (e.g. Bed 14 vs Bed 15). Equipping ward tablets with a barcode scanner where nurses scan the patient's wristband before entering vitals guarantees 100% data integrity and eliminates patient mix-ups.

2. Offline-First Progressive Web App (PWA with Background Sync)

Quarantine wards often have thick concrete or lead-shielded walls with dead Wi-Fi zones. Using service workers and local IndexedDB caching ensures nurses can log vitals completely offline. As soon as the device reconnects to Wi-Fi at the nurse station, readings background-sync automatically.

3. Healthcare Security, Audit Trails & HIPAA/DISHA Privacy

- **Role-Based Field Masking:** Admin staff can see bed occupancy and discharge readiness without exposing sensitive clinical notes or medication histories.
- **Immutable Audit Logging:** Every temperature entry and discharge sign-off records an immutable timestamp, user ID, and IP address to prevent data tampering.
- **Auto-Lock Inactivity Timer:** On shared ward tablets, sessions automatically lock after 3 minutes of inactivity to protect patient privacy.

4. Bluetooth BLE Smart Thermometer Integration

Pairing ward tablets with contactless digital thermometers via the Web Bluetooth API auto-populates readings directly into the app with zero manual typing. At 30 seconds saved per patient, this frees up **~37 minutes per nurse shift across 74 beds**.